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Beyond a Gaussian Denoiser

Residual Learning of Deep CNN for Image Denoising

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PRESENTED BY

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Presentation Roadmap

PART 1

Introduction & Motivation

- Problem statement
- Research gap
- Classical methods
- Mathematical bridge

PART 2

Architecture & Design

- Network architecture
- Residual learning
- RL + BN synergy
- Multi-task extension

PART 3

Data, Training & Results

- Datasets & pipeline
- Evaluation metrics
- Quantitative results
- Visual comparisons

PART 4

Analysis & Conclusion

- Runtime & complexity
- Limitations
- Conclusions
- Future directions

The Image Denoising Problem

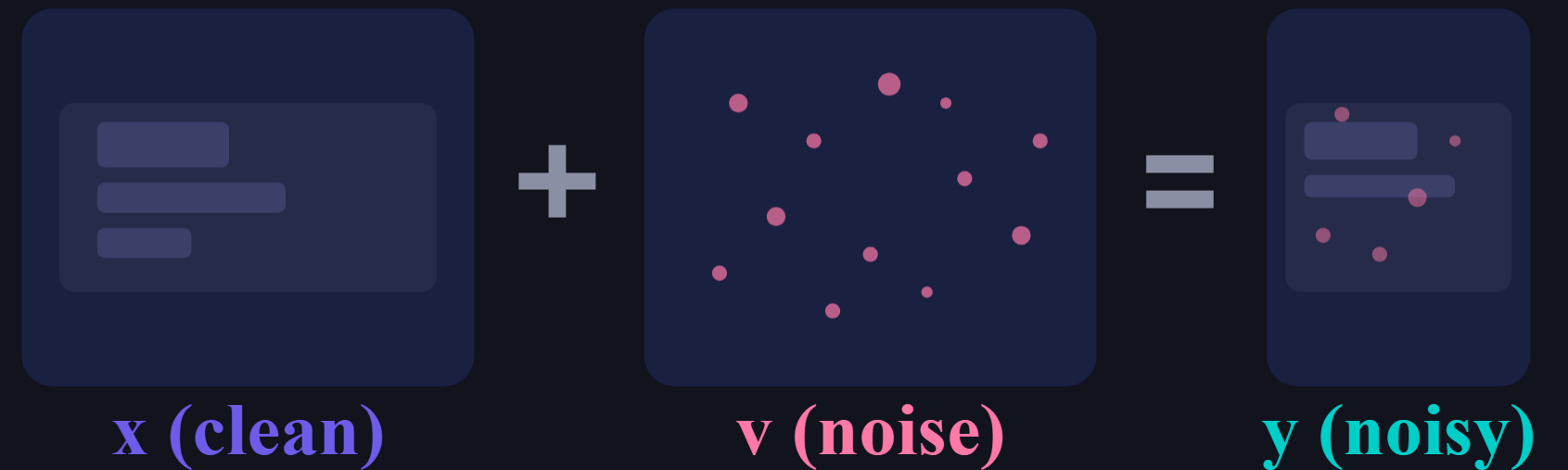
Image Degradation Model

$$\mathbf{y} = \mathbf{x} + \mathbf{v}$$

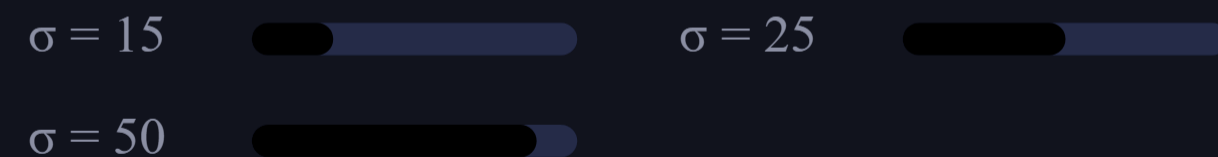
$\mathbf{y}$ Noisy $\mathbf{x}$ Clean $\mathbf{v}$ AWGN (σ)

Applications

Medical imaging — MRI, CT scans
Satellite & remote sensing imagery
Surveillance & security systems
Smartphone & consumer photography
Preprocessing for higher-level vision tasks



Additive White Gaussian Noise (AWGN) model



KEY INSIGHT

Denoising is **ill-posed** — infinitely many clean images could produce the same noisy observation.

Research Gap & Motivation

Limitations of Existing Methods



Computational Cost BM3D, WNNM: complex test-time optimization — minutes per image



Noise-Specific Models CSF, TNRD, MLP: separate model per noise level — no blind denoising



Limited Priors Analysis-model priors cannot capture full image structure

Research Questions

- 1 Can a deep CNN with **residual learning + batch normalization** achieve superior denoising?
- 2 Can a **single model** handle unknown noise levels (blind denoising)?
- 3 Can **one network** handle denoising, SR, and JPEG deblocking?
- 4 Can RL and BN **mutually benefit** each other for Gaussian denoising?

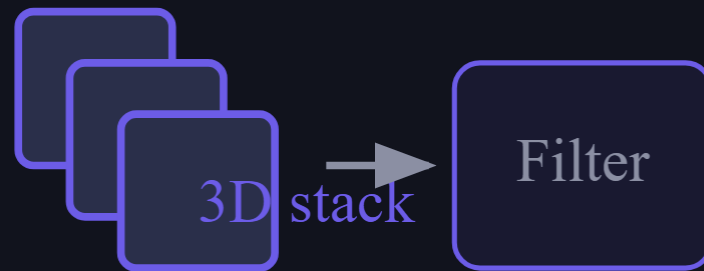


Time to
denoise a
512×512
image

Prior-Based Denoising Approaches

BM3D

2007



Non-local similarity
Collaborative filtering

RF: 49×49

WNNM

2014



Non-local + low-rank
Adaptive weights

RF: 361×361

CSF / TNRD

2014–16



Unrolled optimization
Trainable params

RF: 61×61

DnCNN

2017



Feed-forward CNN · Single model
Blind

RF: 35×35 ✦

From Classical Priors to Learned Features

Classical Bayesian View

$$\hat{x} = \operatorname{argmin} \frac{1}{2} \|y - x\|^2 + \lambda \Phi(x)$$

Data fidelity + Regularization (explicit prior)

DnCNN: Learned Approach

$$\hat{v} = \mathcal{R}(y; \Theta) \Rightarrow \hat{x} = y - \hat{v}$$

No explicit prior — implicitly learned by CNN

Connection: DnCNN Generalizes TNRD

TNRD (1-stage):



⋮ **DnCNN extends by:**

DnCNN:



① $\phi_k \rightarrow \text{ReLU}$

② Much deeper (D layers)

③ + BatchNorm

Most DnCNN parameters represent **learned image priors**, analogous to the analysis prior in TNRD.

PART 2

DnCNN Architecture

Network design, residual
learning, and the RL+BN synergy

ARCHITECTURE

Network Architecture

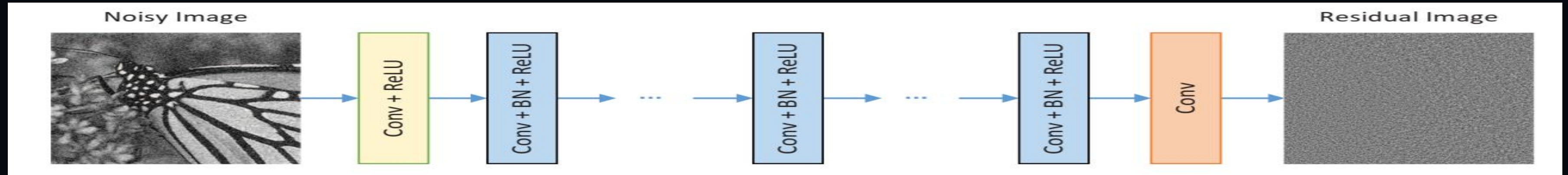


Fig. 1 – DnCNN architecture [Zhang et al., 2017]

Key Design Decisions

Predict **residual noise** $\hat{v}$, not clean image — then $\hat{x} = y - \hat{v}$

No pooling — preserves spatial resolution for pixel-level reconstruction

VGG principle: many small **3×3 filters** with more nonlinearities, fewer params

Batch Normalization in middle layers stabilizes residual learning

FILTERS

3×3

FEATURE Maps

64

POOLING

None

PARAMS

~556K

Residual Learning Strategy

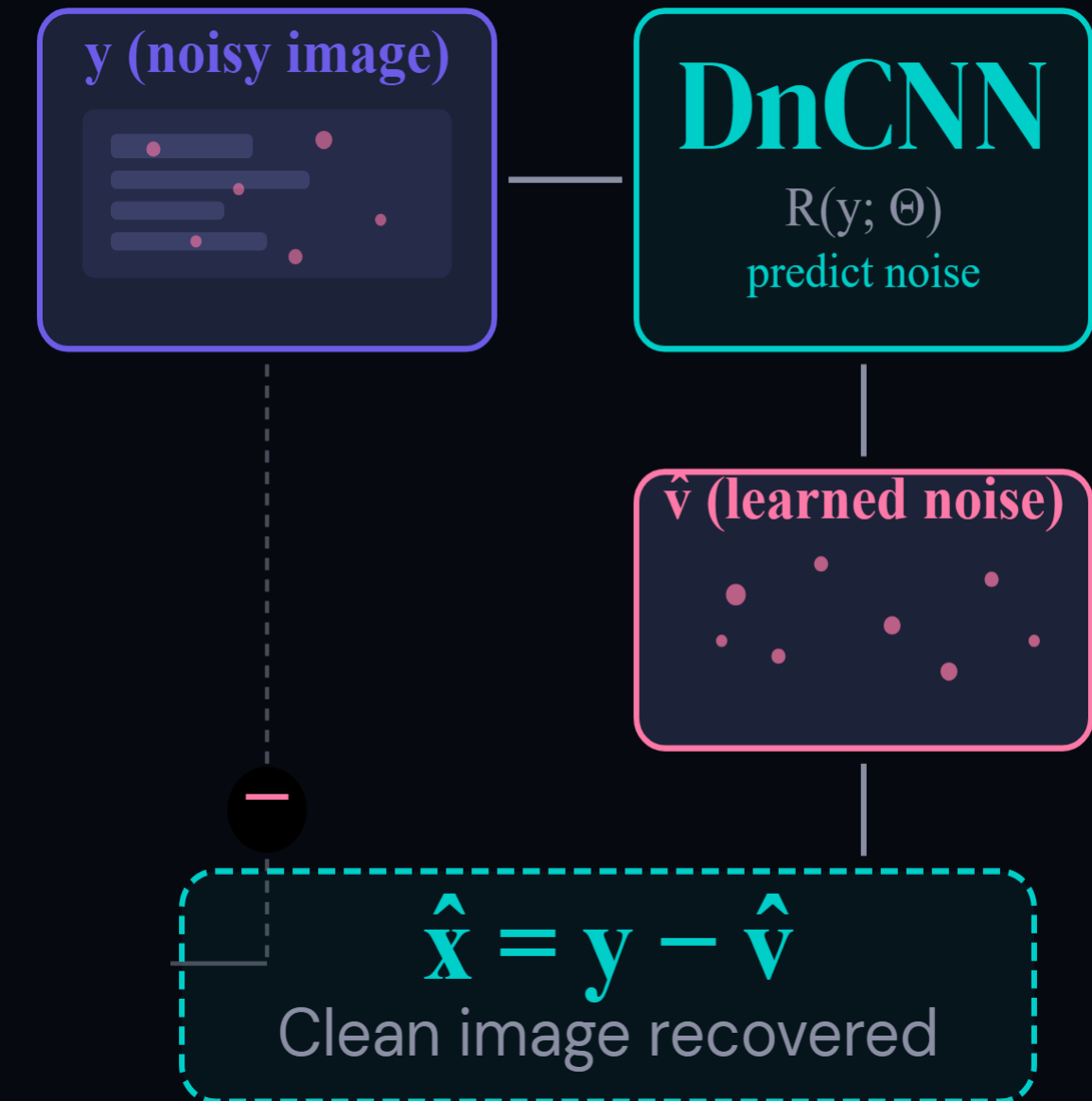
Why Predict Noise Instead of Clean Image?

At low noise, $\mathbf{y} \approx \mathbf{x} \rightarrow$ Learning $F(\mathbf{y}) \approx \text{identity}$ is **hard** $\rightarrow$
Learning $R(\mathbf{y}) = \mathbf{v} \approx \text{small perturbation}$ is **easy**

Loss Function (MSE on Residual)

$$\ell(\Theta) = \frac{1}{2n} \sum_i \|\mathcal{R}(\mathbf{y}_i; \Theta) - (\mathbf{y}_i - \mathbf{x}_i)\|^2$$

Minimizes distance between predicted and actual noise over N training pairs



ARCHITECTURE

RL + BN Synergy

Residual Learning and Batch Normalization are mutually beneficial

RL Benefits from BN

BN alleviates internal covariate shift

Without BN, RL alone cannot match TNRD (28.92 dB)

BN enables stable training with higher learning rates

BN Benefits from RL

Without RL, BN can actually hurt convergence

RL makes hidden layer inputs Gaussian-like

Inputs become less correlated → ideal for BN

Consistent across both SGD and Adam — it's the RL+BN integration, not the optimizer.

Batch Histograms — Why BN Works with RL

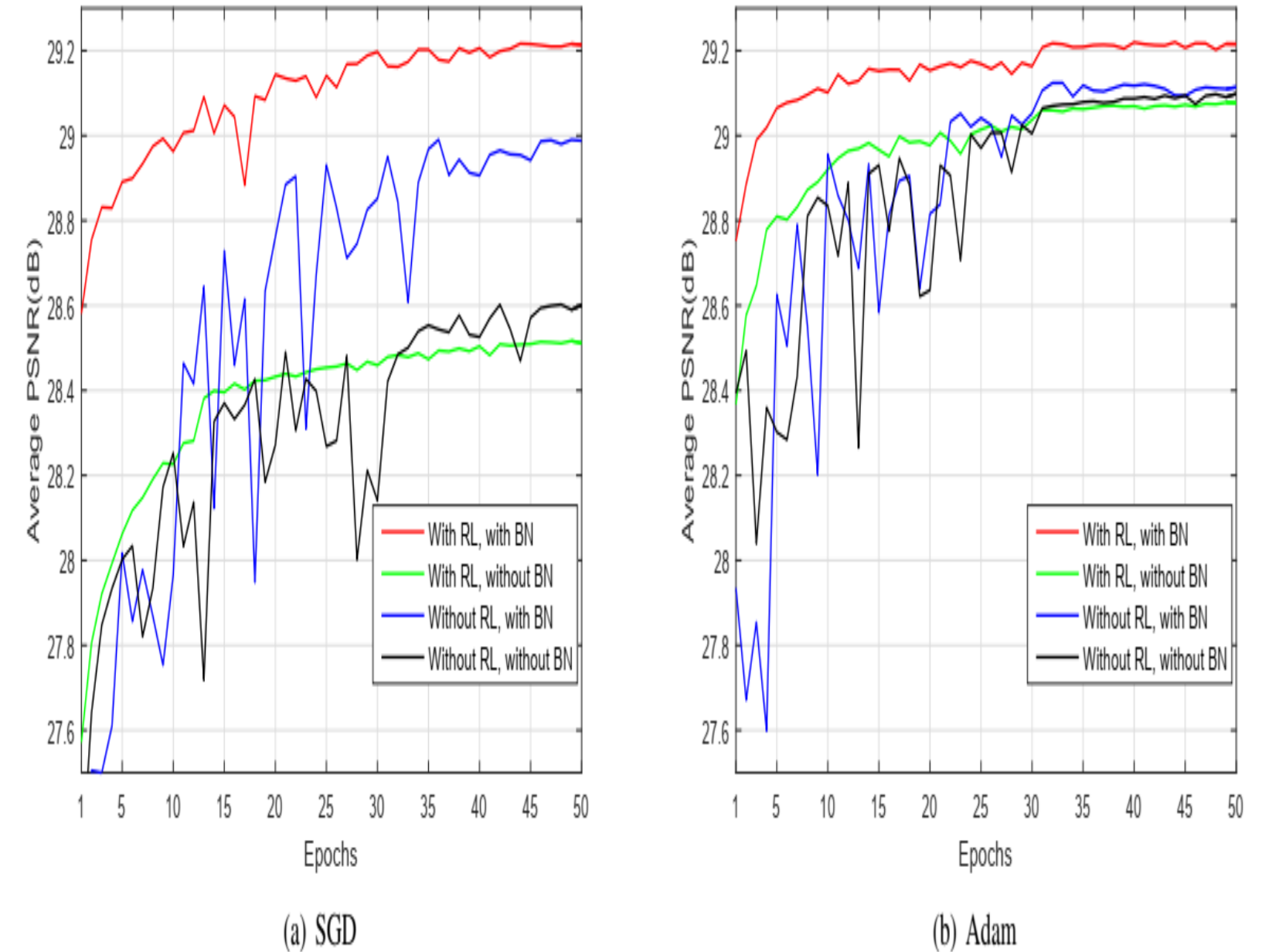


Fig. 2 — RL+BN (red) converges fastest to highest PSNR across SGD and Adam

Network Depth & Receptive Field

$$\text{RF} = (2d + 1) \times (2d + 1)$$

No pooling → each 3×3 conv adds 2 pixels to RF per side

DnCNN-S

Depth 17 · RF 35×35 Known noise level σ

DnCNN-B

Depth 20 · RF 41×41 Blind Gaussian denoising

DnCNN-3

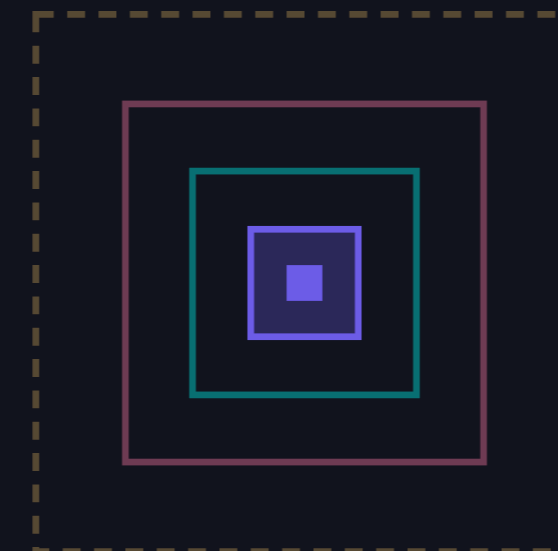
Depth 20 · RF 41×41 Three tasks jointly

Receptive Field Growth

EPLL 36×36

DnCNN 35×35 ✦

BM3D 49×49



3×3 (d=1)

25×25 (d=12)

35×35 (d=17)

41×41 (d=20)

ARCHITECTURE

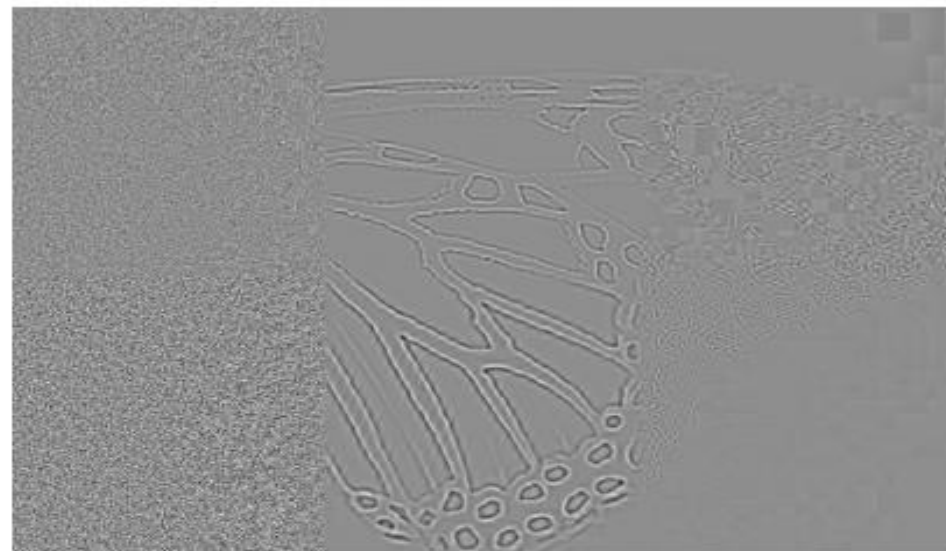
DnCNN-3: One Model, Three Tasks

Unifying Degradation Model

$$y = x + v$$

**Denoising** $v = \text{AWGN } \sigma \in [0, 55]$ **Super-Resolution** $v = \text{HR} - \text{bicubic}(\text{LR}) \times 2, \times 3, \times 4$ **JPEG Deblocking** $v = \text{orig} - \text{compressed QF } 5-99$ **No previous method** had handled all three tasks with one model.

(a) Input Image



(b) Output Residual Image



(c) Restored Image

Fig. 12 – One model handles 6 different degradations simultaneously

PART 3

Data & Training

Datasets, pipeline, optimization,
and hyperparameters

Datasets & Data Pipeline

MODEL	SOURCE	PATCHES	SIZE	NOISE
DnCNN-S	400 imgs (180 ²)	128×1,600	40×40	$\sigma \in \{15, 25, 50\}$
DnCNN-B	400 imgs (180 ²)	128×3,000	50×50	$\sigma \in [0, 55]$
CDnCNN-B	432 BSD color	128×3,000	50×50	$\sigma \in [0, 55]$
DnCNN-3	91+200 BSD	128×8,000	50×50	Mixed tasks

Test Datasets

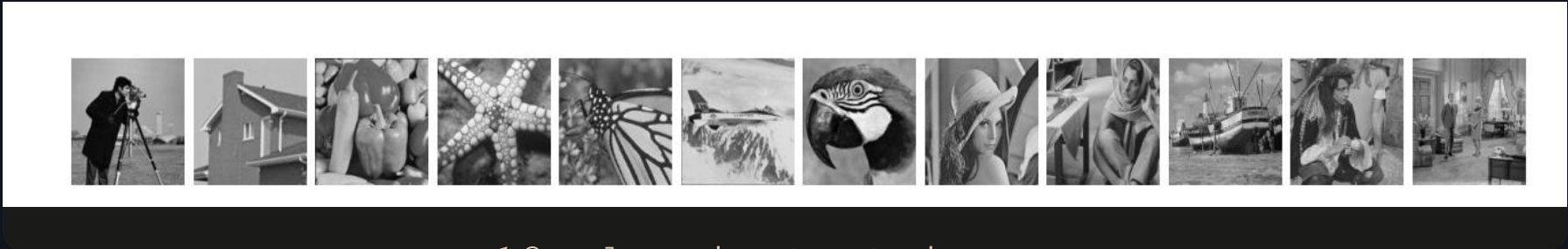
BSD68: 68 natural images (primary benchmark)

12 Classics: Lena, Barbara, Boat, etc.

SR: Set5, Set14, BSD100, Urban100

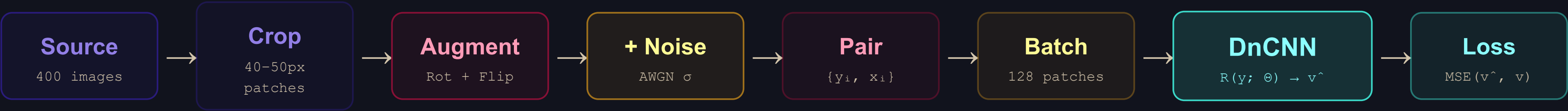
JPEG: Classic5, LIVE1

Data Pipeline



12 classic test images

COMPLETE TRAINING PIPELINE



Training Configuration

LOSS FUNCTION

MSEOn residual $\|R(y;\Theta)-(y-x)\|^2$

OPTIMIZER

SGDMomentum 0.9 · WD $1e-4$

LEARNING RATE

 $1e^{-1} \rightarrow 1e^{-4}$

Exponential decay / 50 ep.

BATCH SIZE

128

Patch mini-batch SGD

INITIALIZATION

He et al.

Variance scaling for ReLU

FRAMEWORK

MatConvNet

MATLAB-based CNN lib

CPU

Intel i7-5820K CPU @ 3.30GHz

GPU

Nvidia Titan X GPU

Learning Rate Schedule



Training Time

DnCNN-S

~6h

DnCNN-B

~1 day

DnCNN-3

~3 days

PART 3 (CONT.)

Experimental Results

Quantitative benchmarks, visual
comparisons, and multi-task
evaluation

RESULTS

Evaluation Metrics

PSNR (Peak Signal-to-Noise Ratio)

$$\text{PSNR} = 10 \cdot \log_{10}(\text{MAX}^2 / \text{MSE}) \text{ dB}$$

Higher = better · Standard metric for low-level vision

0.5 dB improvement is considered **significant**

Pixel-wise — may not match visual perception

SSIM (Structural Similarity Index)

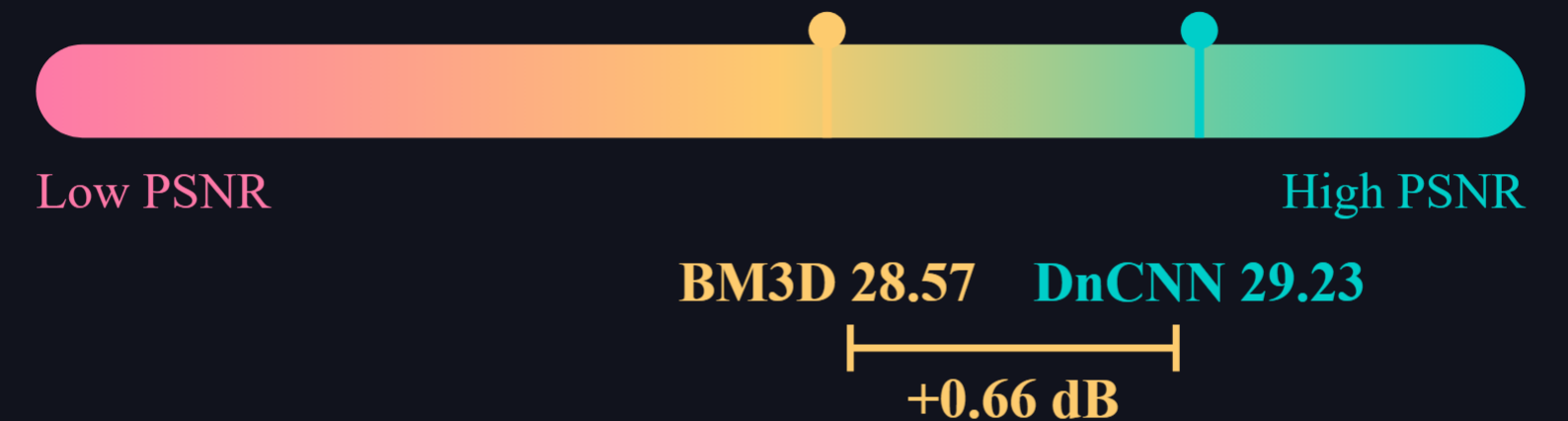
$$\text{SSIM}(x, y) = f(\text{luminance}, \text{contrast}, \text{structure})$$

Range [0, 1]: 1 = identical

Considers structural information, not just pixel error

Better correlates with human visual perception

What PSNR Means in Practice



Theoretical bound over BM3D: ~0.7 dB

Levin & Nadler, 2011

Before DnCNN, few methods could beat BM3D by **>0.3 dB** on average. DnCNN-S achieves **+0.6 dB** — approaching the theoretical limit.

RESULTS

Gaussian Denoising — BSD68

Average PSNR (dB) comparison · Table II in paper

METHOD	$\Sigma = 15$	$\Sigma = 25$	$\Sigma = 50$
BM3D <i>[Dabov '07]</i>	31.07	28.57	25.62
WNNM <i>[Gu '14]</i>	31.37	28.83	25.87
EPLL <i>[Zoran '11]</i>	31.21	28.68	25.67
MLP <i>[Burger '12]</i>	—	28.96	26.03
TNRD <i>[Chen '16]</i>	31.42	28.92	25.97
DnCNN-S	31.73	29.23	26.23
DnCNN-B (blind)	31.61	29.16	26.23

+0.6 dB

DnCNN-S over BM3D at all noise levels

DnCNN-B (blind, unknown σ) still beats all noise-specific competitors

RESULTS

Per-Image Analysis — $\sigma = 25$

IMAGE	BM3D	WNNM	TNRD	DnCNN-S	Δ BM3D
C.man	29.45	29.64	29.72	30.18	+0.73
House	32.85	33.22	32.53	33.06	+0.21
Barbara	30.71	31.24	29.41	30.00	-0.71
Parrot	28.93	29.15	29.18	29.43	+0.50
Avg (12)	29.97	30.26	30.06	30.44	+0.47

Strengths

Best on most images at all σ
+0.2–0.6 dB on textured images
Sharp edges + clean smooth regions

Where It Struggles

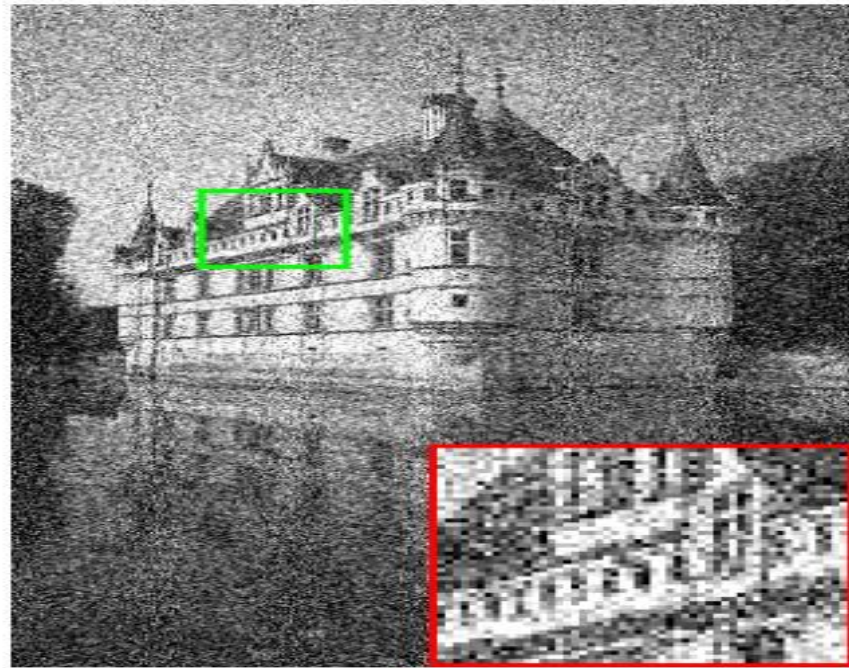
Barbara/House: repetitive patterns
Non-local methods (WNNM) win here
DnCNN lacks non-local prior

Irregular textures → DnCNN wins
Regular patterns → non-local wins

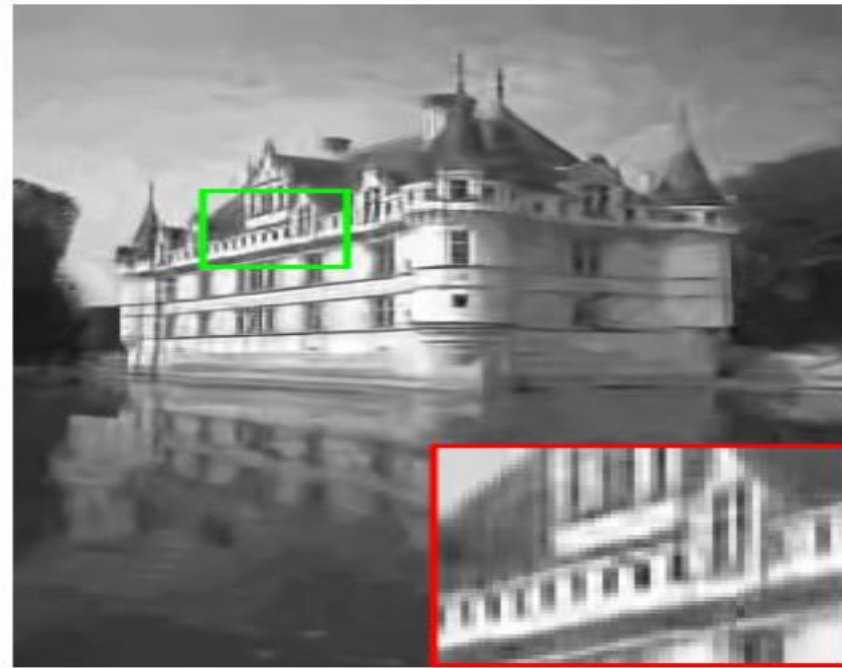
RESULTS

Visual Quality Comparison

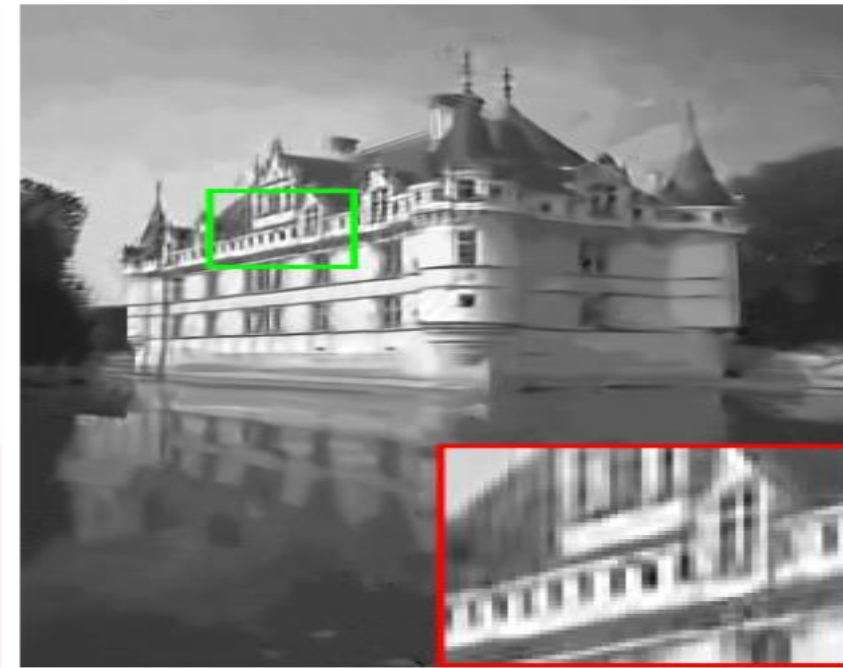
Qualitative analysis at $\sigma = 50$



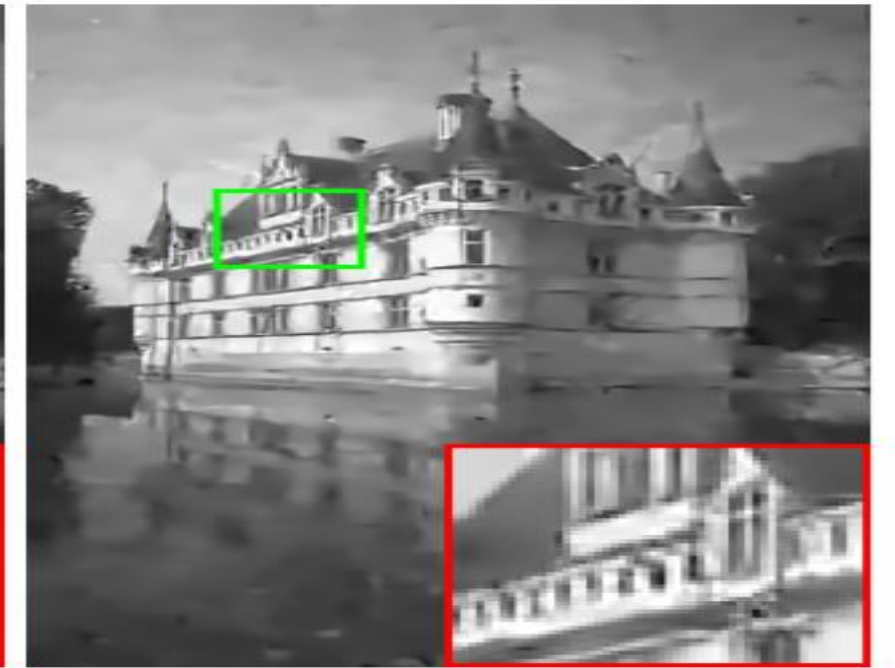
(a) Noisy / 14.76dB



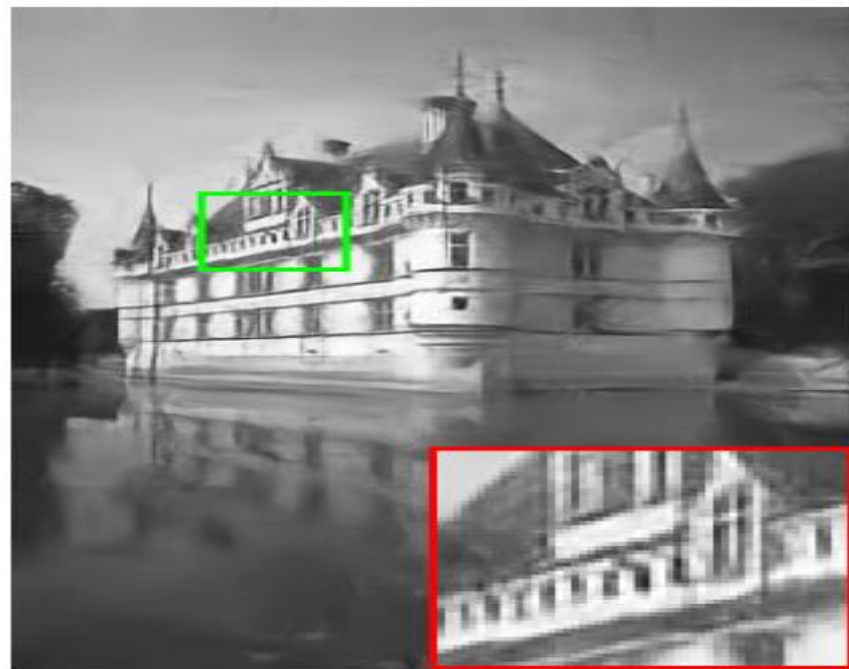
(b) BM3D / 26.21dB



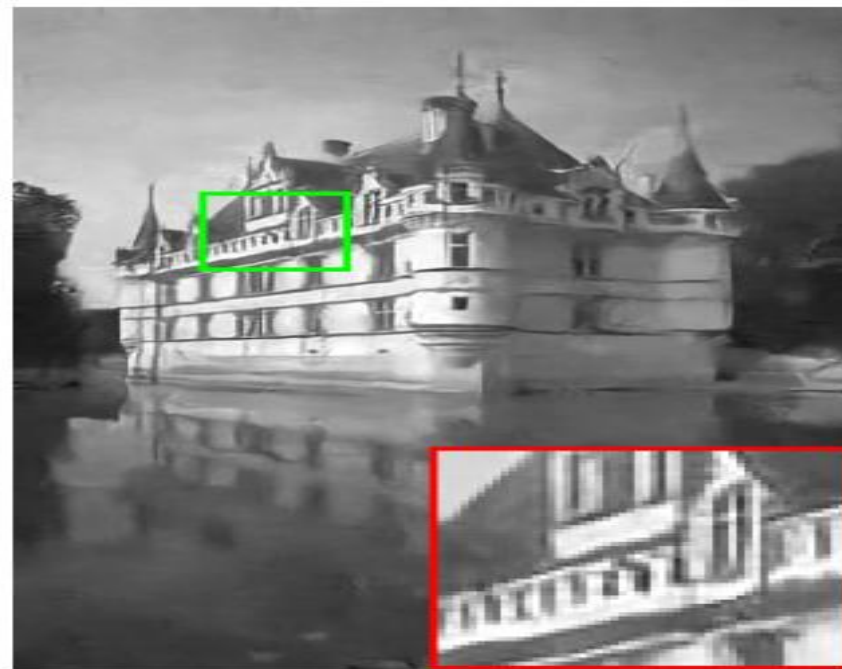
(c) WNNM / 26.51dB



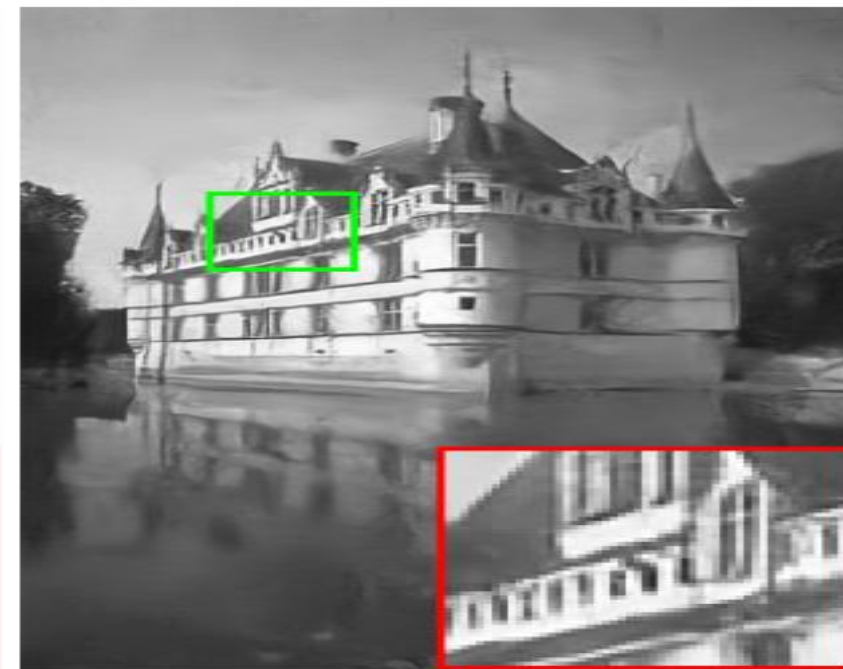
(d) EPLL / 26.36dB



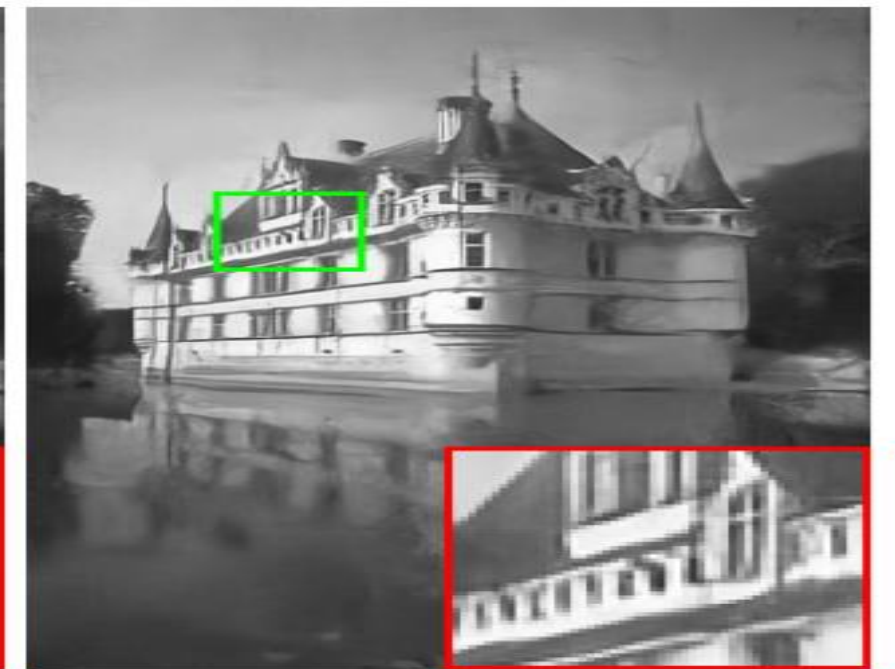
(e) MLP / 26.54dB



(f) TNRD / 26.59dB



(g) DnCNN-S / 26.90dB



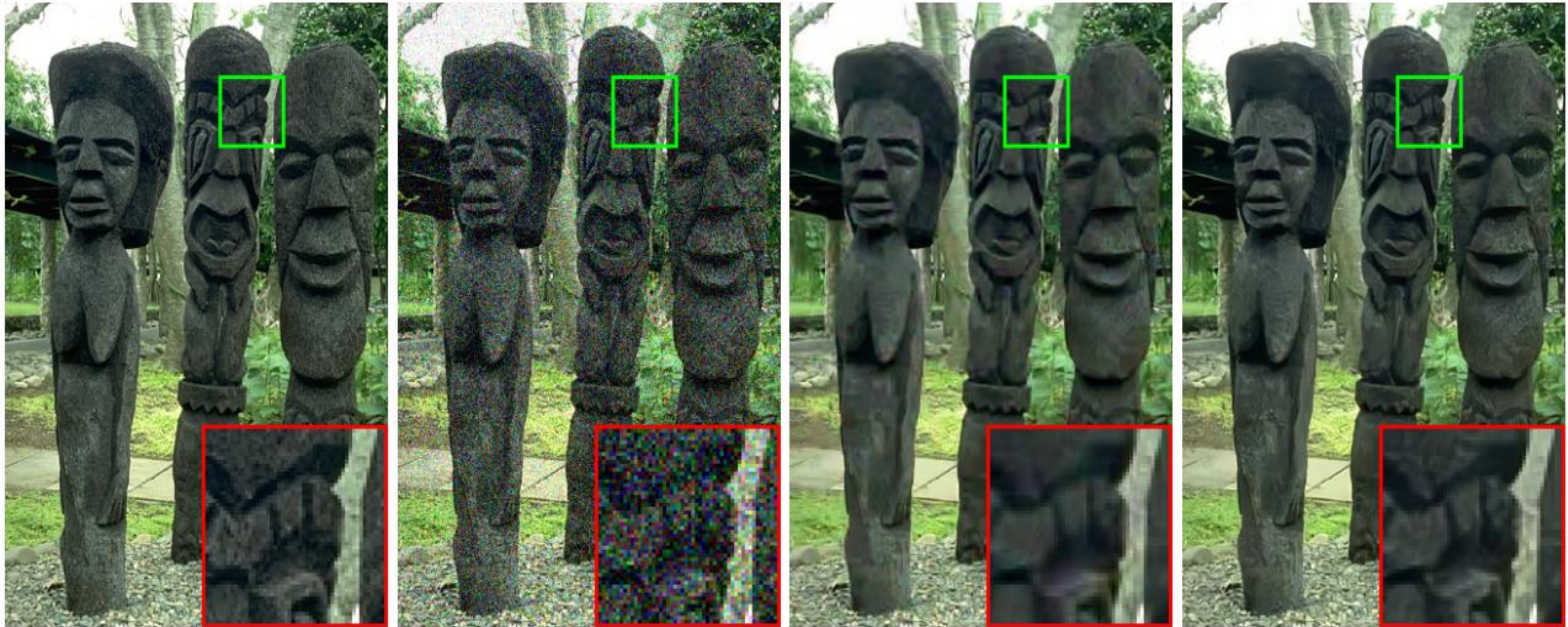
(h) DnCNN-B / 26.92dB

Fig. 4 – Castle at $\sigma=50$: BM3D over-smooths, TNRD has artifacts, DnCNN-S/B preserve edges + clean smooth areas

RESULTS

Color Image Denoising

CDnCNN-B vs CBM3D — $\sigma = 35$



(a) Ground-truth

(b) Noisy / 17.25dB

(c) CBM3D / 25.93dB

(d) CDnCNN-B / 26.58dB

Fig. 6 — CDnCNN-B produces more natural color and sharper detail than CBM3D

RESULTS

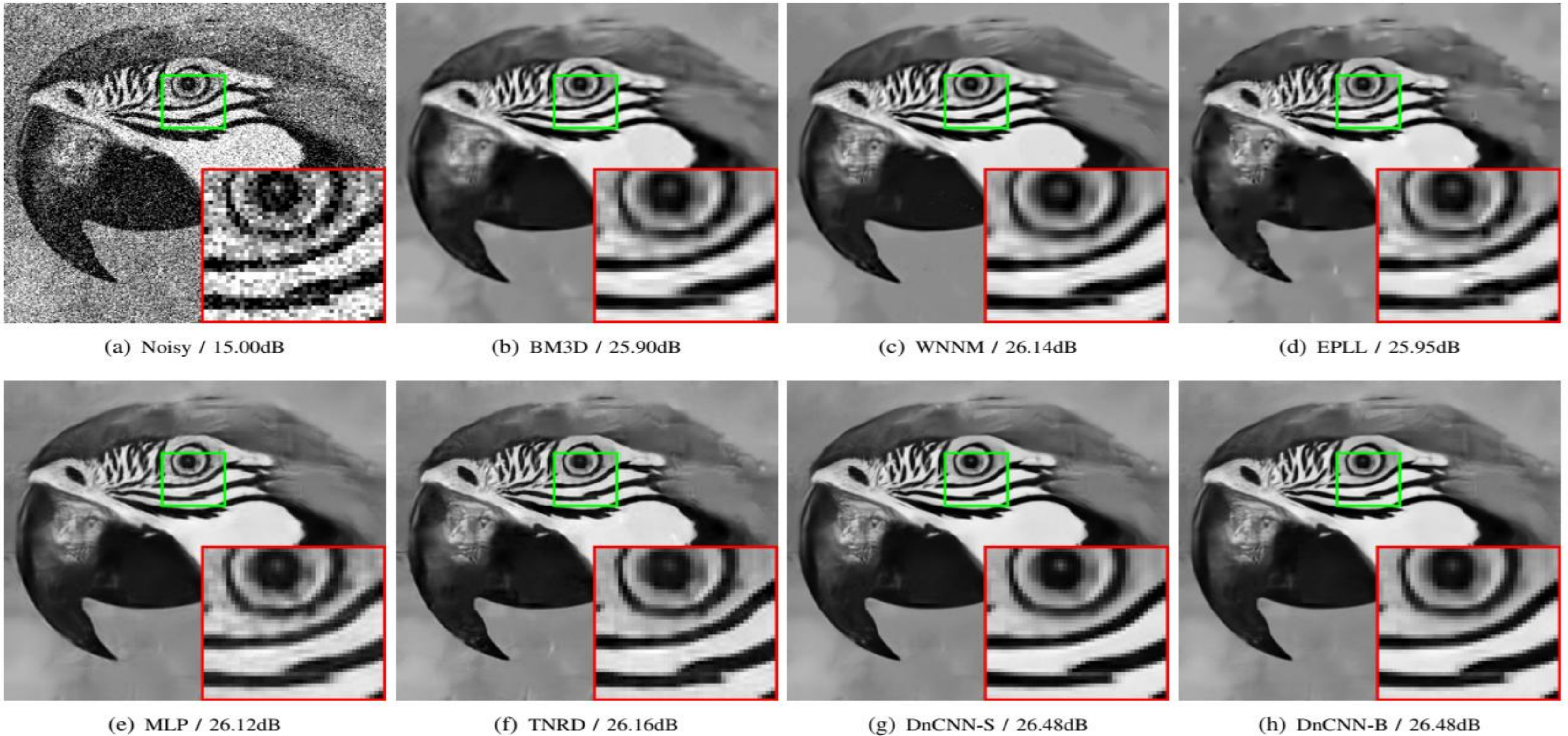
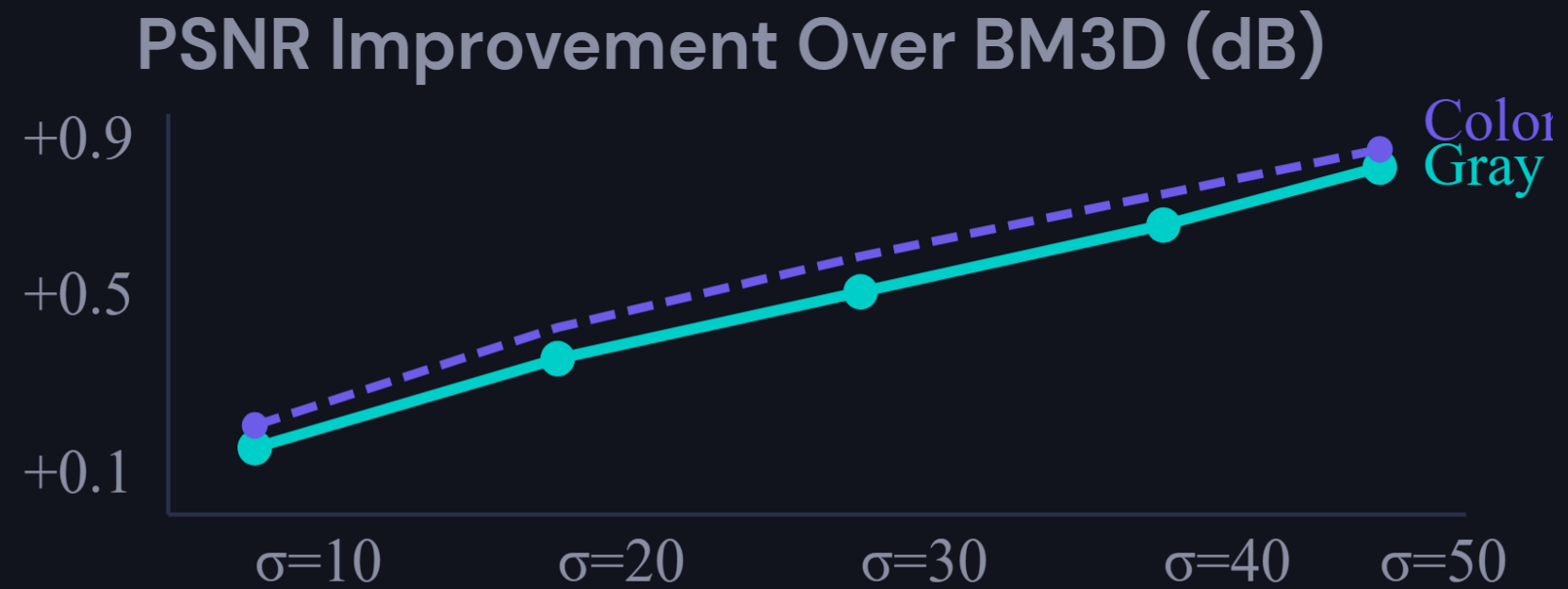
Visual Quality — Parrot ($\sigma = 25$)

Fig. 5 — DnCNN preserves fine detail around the eye while keeping smooth regions clean

RESULTS

Blind & Multi-Task Performance

DnCNN-B: Blind Denoising



No noise estimation step needed

Improvement **increases** with noise level

Single model simplifies deployment

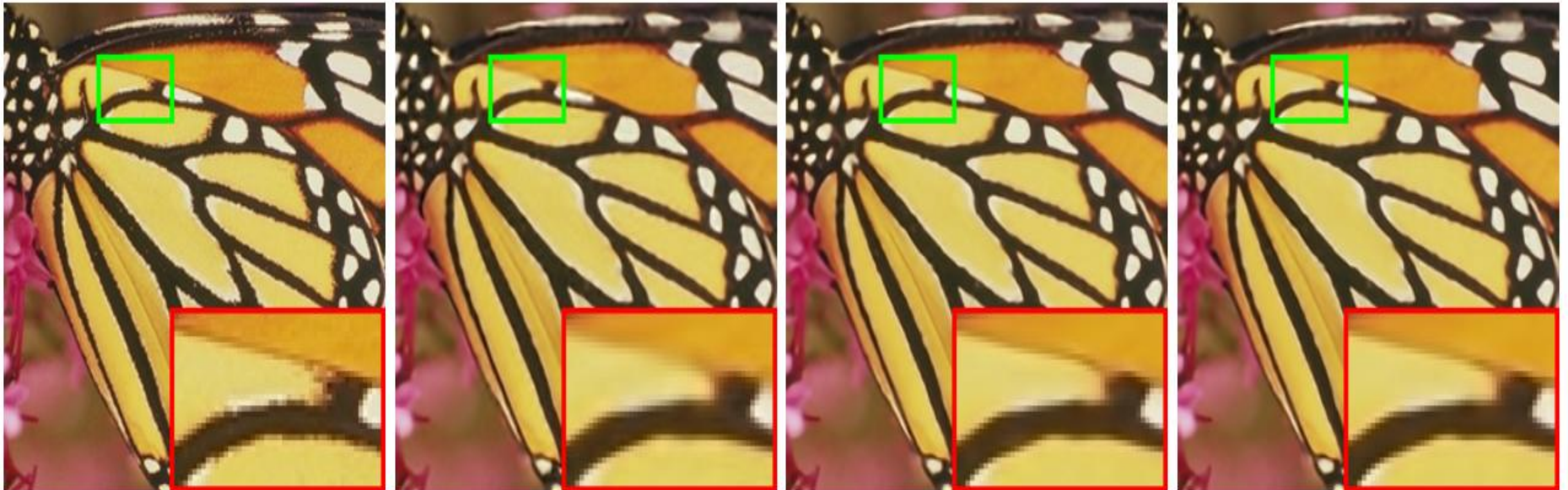
DnCNN-3: One Model, Three Tasks

Single model matches or beats task-specific SOTA across all three tasks.

RESULTS

Super-Resolution — DnCNN-3

Single model handles $\times 2$, $\times 3$, $\times 4$ upscaling



(a) Ground-truth

(b) TNRD / 28.91dB

(c) VDSR / 29.95dB

(d) DnCNN-3 / 30.02dB

Fig. 9 — SR $\times 3$: DnCNN-3 (30.02 dB) outperforms TNRD (28.91 dB) and matches VDSR (29.95 dB)

RESULTS

JPEG Deblocking — DnCNN-3

Removing compression artifacts with a single multi-task model



Fig. 11 — JPEG QF 10: DnCNN-3 (29.70 dB) removes blocking artifacts better than AR-CNN and TNRD

PART 4

Critical Analysis

Runtime, complexity, model
compression, and limitations

COMPLEXITY

Runtime & Computational Efficiency

METHOD	256 ²	512 ²	1024 ²
BM3D	0.65s	2.85s	11.89s
WNNM	203s	773s	2536s
EPLL	25.4s	45.5s	422s
TNRD (GPU)	0.010s	0.032s	0.116s
DnCNN-S (GPU)	0.014s	0.051s	0.200s
DnCNN-B (GPU)	0.016s	0.060s	0.235s

DnCNN on GPU — 60ms for 512×512 (real-time capable)

WNNM — 773s for same image (12,000× slower)

Feed-forward inference → **predictable latency**

All ops are standard conv/BN → **cuDNN accelerated**

60ms

512×512 GPU

12K×

vs WNNM

556K

parameters

COMPLEXITY

Model Complexity & Optimization

Parameter Breakdown

Layer 1: $3 \times 3 \times 1 \times 64$	576
Layers 2-16: $15 \times (3 \times 3 \times 64 \times 64)$	552,960
Layer 17: $3 \times 3 \times 64 \times 1$	576
BN params: $15 \times 2 \times 64$	1,920

Total ~556K params

Deployment Advantages

- Feed-forward: fixed computation per pixel
- GPU-friendly: standard conv/BN ops
- cuDNN v5 acceleration supported
- Predictable latency, no iteration

Potential Optimization Strategies

 **Pruning** Remove redundant filters — 30-50% param reduction

 **Quantization** INT8/FP16 — 2-4× memory reduction

 **Knowledge Distillation** Shallower student from DnCNN teacher

 **Depthwise Separable Conv** Replace standard convs — ~8-9× FLOP reduction

Limitations & Failure Cases

Non-Local Patterns

Local RF cannot exploit long-range self-similarity.
WNNM/BM3D win on Barbara (-0.71 dB vs BM3D).

AWGN Assumption

Real-world noise is signal-dependent, spatially varying.
AWGN is a simplification that limits practical applicability.

No Perceptual Loss



High PSNR



Perceptual quality

MSE optimizes PSNR, not visual quality. GANs and perceptual losses (SRGAN) showed these can diverge.

Fixed Architecture

No attention

No multi-scale

No adaptation

Later works (DRUNet, SwinIR, Restormer) showed attention + multi-scale significantly improve results.

Key Contributions & Impact

1 Residual Learning for Denoising

First to predict $\hat{v}$ (noise) instead of $\hat{x}$ (clean). Connected theoretically to TNRD.

2 RL + BN Synergy

RL makes inputs Gaussian-like → ideal for BN. BN stabilizes RL. Together: fastest convergence, best quality.

3 SOTA Performance

+0.6 dB over BM3D · Approaching theoretical bound ·
60ms on GPU

4 Blind & Multi-Task

One model for unknown σ (DnCNN-B). One model for 3 tasks (DnCNN-3). Both beat task-specific SOTA.

DnCNN is cited **6,000+** times and established **residual learning + BN** as the standard denoising paradigm.

FUTURE WORK

Research Directions & Legacy

 Attention

Non-Local Attention

Self-attention blocks to capture long-range dependencies DnCNN misses

→ Non-local Neural Networks, NLRN

 Losses

Perceptual & GAN

VGG perceptual loss + adversarial training for visual realism over PSNR

→ SRGAN, ESRGAN

 Noise

Real-World Noise

Signal-dependent, spatially varying, sensor-specific noise models

→ FFDNet, CBDNet, SCUNet

 Efficiency

Lightweight Models

NAS, depthwise convolutions, distillation for mobile/edge deployment

→ ECBSR, IMDN

 Architecture

Transformers & U-Nets

Multi-scale + attention architectures built on DnCNN's foundations

→ SwinIR, Restormer, DRUNet

 Self-Supervised

No Clean Images

Train denoiser without ground truth clean images

→ Noise2Noise, Noise2Void

REFERENCES

Key References

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Thank You

Questions & Discussion

Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising

Zhang, Zuo, Chen, Meng & Zhang — IEEE TIP, 2017